

Data quality rules using dimensions

INTRODUCTION TO DATA QUALITY



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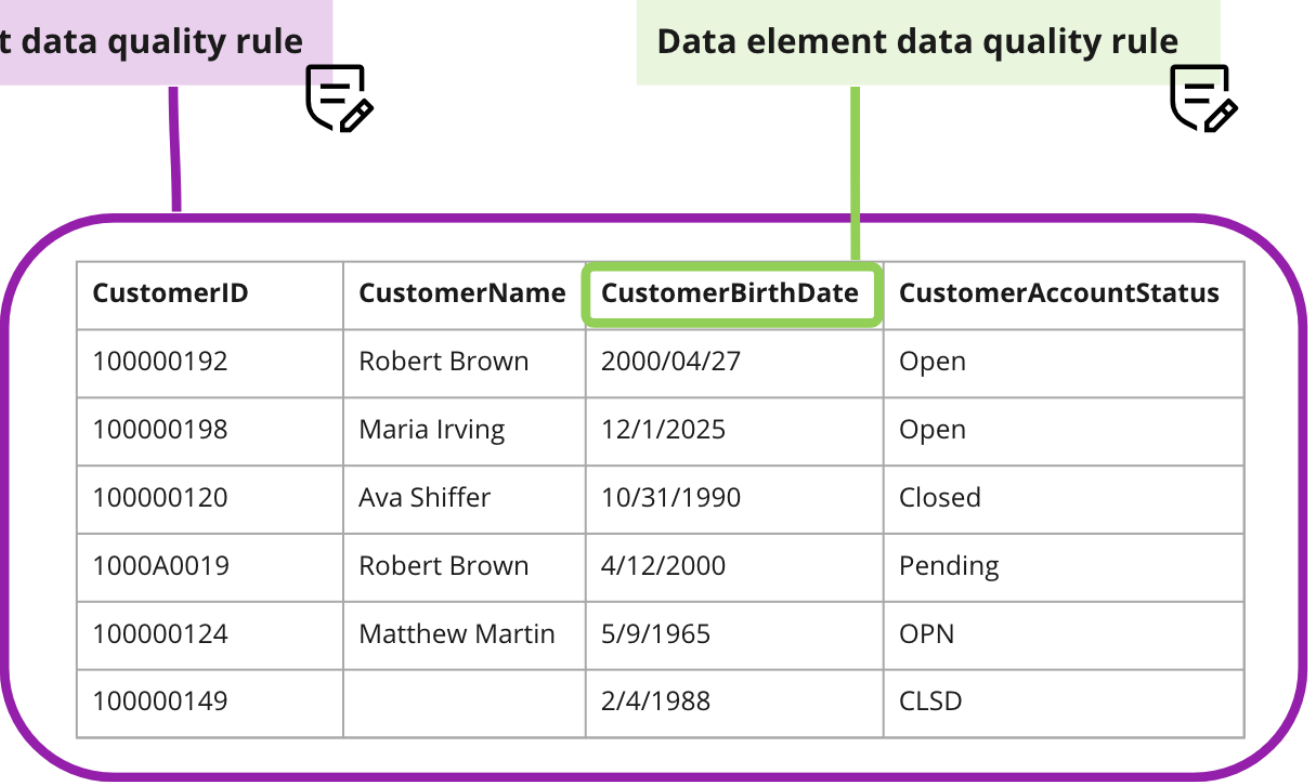
Data quality rules

Data quality rule: type of business rule that validates whether data meets business requirements

- Can be defined at:
 - Dataset level
 - Data element level

Dataset data quality rule

Data element data quality rule



CustomerID	CustomerName	CustomerBirthDate	CustomerAccountStatus
100000192	Robert Brown	2000/04/27	Open
100000198	Maria Irving	12/1/2025	Open
100000120	Ava Shiffer	10/31/1990	Closed
1000A0019	Robert Brown	4/12/2000	Pending
100000124	Matthew Martin	5/9/1965	OPN
100000149		2/4/1988	CLSD

Completeness data quality rule

Dataset data quality rule: All expected records from the source must be loaded in the target table.

Count of records in SourceCustomerTable	Count of records in TargetCustomerTable
5,090,203	5,090,203



Data element data quality rule: All records must have a Customer ID, Customer Name, and Customer Account Type populated.

CustomerID	CustomerName	CustomerAccountType
100000192	Robert Brown	Loan
100000198	Maria Irving	
100000120	Ava Shiffer	Credit Card
100000192	Robert Brown	
100000124	Matthew Martin	Deposit
100000149		Loan

Timeliness data quality rule

Dataset data quality rule: All records in the customer dataset must be loaded by the 9:00 am.

Customer table SLA = 9:00 AM

CustomerID	TableLoadDateTime
100000192	01-01-2023 11:07 am
100000198	01-01-2023 11:07 am
100000120	01-01-2023 11:07 am

Data element data quality rule: All records must have a tax ID populated by first time the customer's account status is "Open".

Customer table SLA = 9:00 AM

CustomerID	CustomerName	CustomerTaxID	AccountStatus
100000192	Robert Brown	NULL	Pending
100000198	Maria Irving	xxx-xx-2039	Open
100000120	Ava Shiffer	NULL	Open
100000192	Robert Brown	xxx-xx-5637	Open
100000124	Matthew Martin	NULL	Pending

Validity data quality rule

Data element data quality rule:

- All records must have a Birth Date value in the format MM/DD/YYYY and the value must be in the past.
- All records must have an Account Status of Open, Closed, or Pending.

CustomerID	CustomerName	CustomerBirthDate	CustomerAccountStatus
100000192	Robert Brown	2000/04/27 ✕	Open
100000198	Maria Irving	12/1/2025	Open
100000120	Ava Shiffer	10/31/1990	Closed
1000A0019	Robert Brown	4/12/2000	Pending
100000124	Matthew Martin	5/9/1965	OPN ✕
100000149		2/4/1988	CLSD ✕

Consistency data quality rule

Dataset data quality rule: The count of records loaded today must be within +/- 5% of the count of records loaded yesterday.

Count of records in TargetCustomerTable	Record count difference from previous day	TargetCustomerTable Load Date
10,000,000	4,909,797 	01-02-2023 08:34 am
5,090,203	75 	01-01-2023 11:07 am
5,090,128	1 	12-31-2023 08:38 am

Data element data quality rule: All Customer ID values in the AccountTable must also be present in the CustomerTable.

AccountTableCustomerID	CustomerTableCustomerID
108394858	108394858
192039482	192039482
203475849	NULL 
2930485953	NULL 
102832748	102832748

Accuracy data quality rule

Data element data quality rule: All records in the CustomerTable must have accurate Customer Name, Customer Birthdate, and Customer Address fields when compared to the Tax Form.

CustomerName	CustomerBirthDate	CustomerAddress	CustomerCity	CustomerState	CustomerZip
Ava Shiffer	10/31/1990	910 Quality St	Washington	WA	20008

Tax Form

Name: Ava Shiffer Birthdate: 10/30/1990

Address: 910 Quality St

City: Washington State: DC

Zip: 20008



Uniqueness data quality rule

Data element data quality rule:

- All records must have a unique Customer ID.
- All records must have a unique combination of customer name, customer birth date, and customer address fields.

CustomerID	CustomerName	CustomerBirthDate	CustomerAddress
100000192	Robert Brown	4/12/2000	703 Data Drive
1000001989	Maria Irving	12/1/2025	31 Duplicate Way
100000120	Ava Shiffer	10/31/1990	910 Quality St
100000190	Robert Brown	4/12/2000	703 Data Drive

Let's practice!

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Data profiles

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What is data profiling?

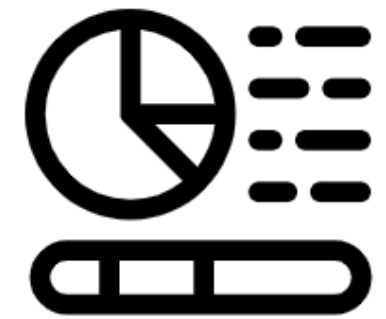
Data profiling: The activity of running statistics on a data set to better understand the data and field dependencies

Examples:

- How many records are in the data set?
- What are the min and max values for a particular data element?
- How many records have a particular data element populated?
- When column A is populated, what other columns are also populated?

Data Profile

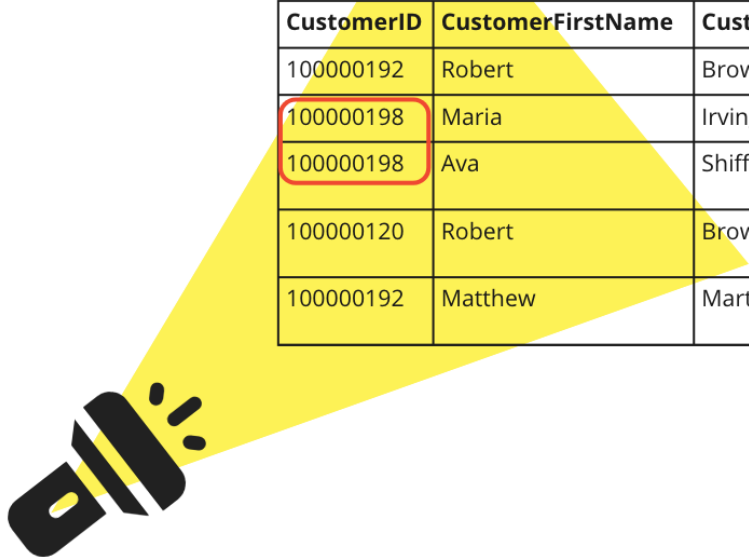
- **Min value**
- **Max value**
- **# of nulls**
- **Most common value**



Importance of data profiling

Data profiling:

- Confirms what you already know
- Reveals what you don't know
- Identifies data quality issues
- Aids in writing better data quality rules



CustomerID	CustomerFirstName	CustomerLastName	CustomerBirthDate	CustomerAccountType
100000192	Robert	Brown	4/12/2000	Loan
100000198	Maria	Irving	12/1/2025	Deposit
100000198	Ava	Shiffer	10/31/1990	Credit Card
100000120	Robert	Brown	4/12/2000	Deposit
100000192	Matthew	Martin	5/9/1965	Deposit

What does a data profile look like?

CustomerID	CustomerFirstName	CustomerLastName	CustomerBirthDate	CustomerAccountType
100000192	Robert	Brown	4/12/2000	Loan
100000198	Maria	Irving	12/1/2025	Deposit
100000198	Ava	Shiffer	10/31/1990	Credit Card
100000120	Robert	Brown	4/12/2000	Deposit
100000192	Matthew	Martin	5/9/1965	Deposit



**Data
Profile**

Customer ID data profile

CustomerID
100000192
100000198
100000198
100000120
100000192

- **Populated for all 15,430 records**
- **Numeric values with 11 characters**
- **Min value = 000000000001**
- **Max value = 999999999999**



Customer Name data profile

- Populated for 98% of records
- String of text
- Min Length = 1
- Max Length = 20
- 95% of records have a Customer First Name and Customer Last Name
- Most common value = Jennifer

CustomerFirstName	CustomerLastName
Robert	Brown
Maria	Irving
Ava	Shiffer
Robert	Brown
Matthew	Martin

- Populated for 96% of records
- String of text
- Min Length = 1
- Max Length = 30
- 95% of records have a Customer First Name and Customer Last Name
- Most common value = Smith



Customer Birth Data data profile

CustomerBirthDate
4/12/2000
12/1/2025
10/31/1990
4/12/2000
5/9/1965

- **Populated for 76% of records**
- **98% of records conform to the date format MM/DD/YYYY**
- **Min value = 01/01/1900**
- **Max value = 99/99/9999**
- **75% of records have a Birth Date and a First and Last Name populated**



Customer Account Type data profile

CustomerAccountType
Loan
Deposit
Credit Card
Deposit
Deposit

- Populated for 99% of records
- 1% are null
- 22% have a value of Loan
- 34% have a value of Deposit
- 30% have a value of Loan and Deposit
- 9% have a value of Credit Card
- 4% have other various values.



**Data
Profile**

Using a data profile in data quality

Field Name	Valid Values
CustomerID	11 numeric characters
CustomerFirstName	Text string 1-20 characters
CustomerLastName	Text string 1-30 characters

- All CustomerID values must be 11 numeric characters.
- All CustomerFirstName values must be 1 - 20 character string of text.
- All CustomerLastName values must be 1 - 30 character string of text.

Field Name	Valid Values
CustomerBirthDate	MM/DD/YYYY formatted dates between 01/01/1900 and 99/99/9999
CustomerAccountType	Loan, Deposit, Loan and Deposit, Credit Card

- All CustomerBirthDate values must be in the MM/DD/YYYY format and between 01/01/1900 and 99/99/9999.
- All CustomerAccountType values must be Loan, Deposit, Loan and Deposit, or Credit Card.

Let's practice!

INTRODUCTION TO DATA QUALITY

Metadata and data quality

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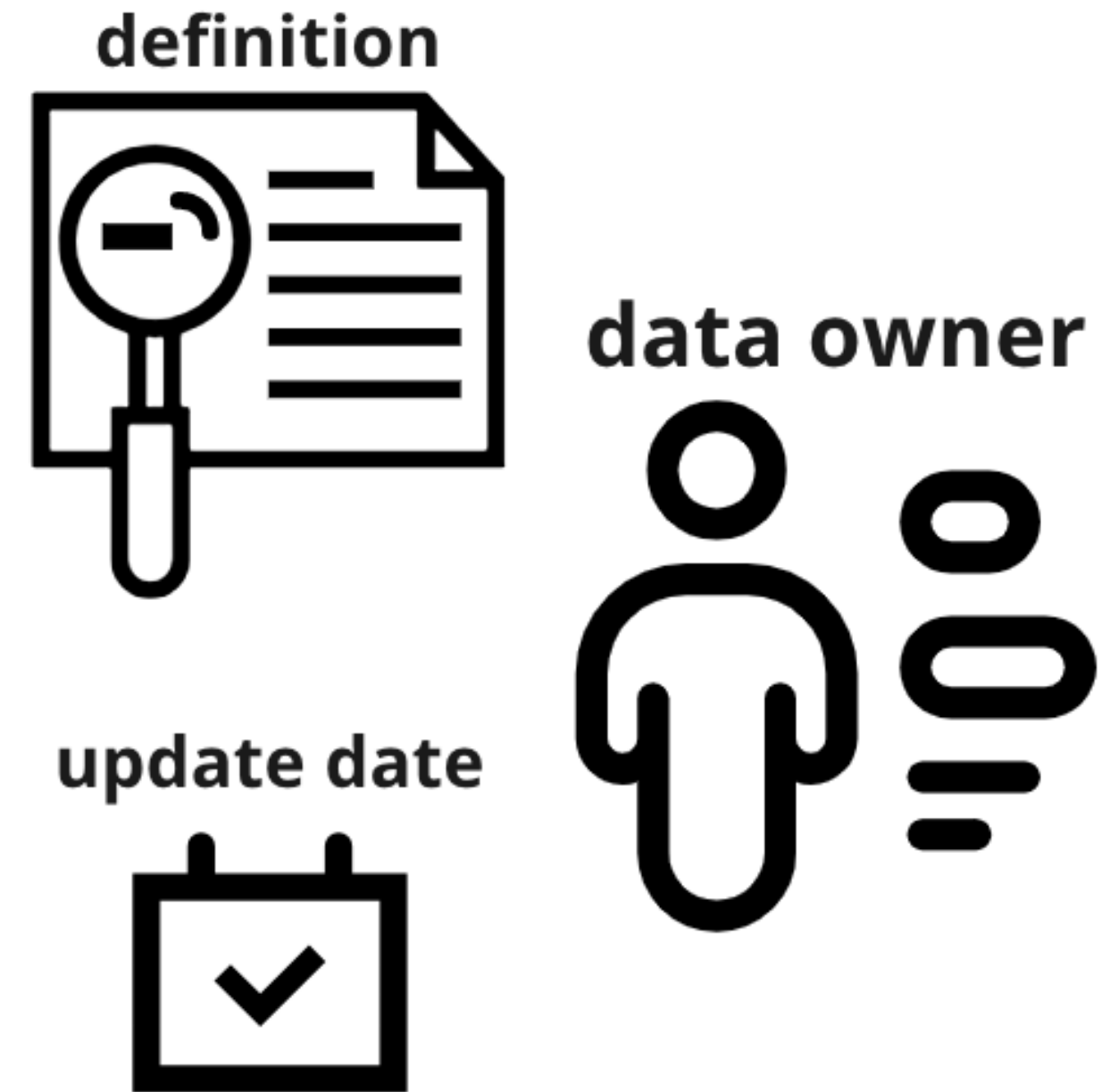
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What is metadata?

Metadata: data about data, or attributes that describe data

- Used to organize and understand datasets and data elements
- Used in the data quality process to determine the:
 - definition of a field
 - owner of a field
 - field's last update date



Metadata examples

Metadata can be found in a data dictionary.

Examples:

- Business field name
- Business definition
- Data owner
- Technical physical field name

Metadata

SEARCH: Customer First Name

Business Field Name: Customer First Name

Business Definition: The first name of the customer which is provided in the tax form and confirmed by the customer.

Data Owner: Rita Walker

Last Updated Date: 01/01/2023 

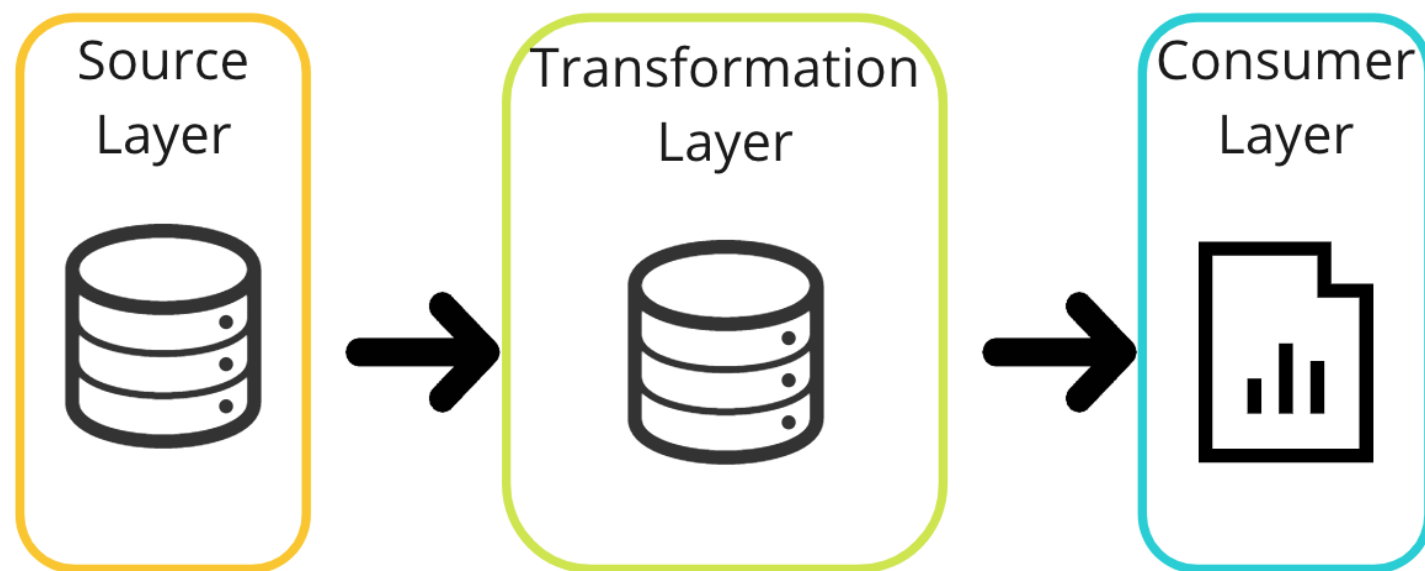
Technical Physical Field Name: CustomerFirstName

Technical Physical Table Name: CustomerTable

What is data lineage?

Data lineage: A representation of how data moves in a pipeline, from where the data is entered in the source through each step in the data pipeline, until it is consumed.

- Each layer has its own metadata
- Used in the data quality process to determine where to implement a data quality rule



Data lineage example

Customer Source Tables



Customer
Staging Table



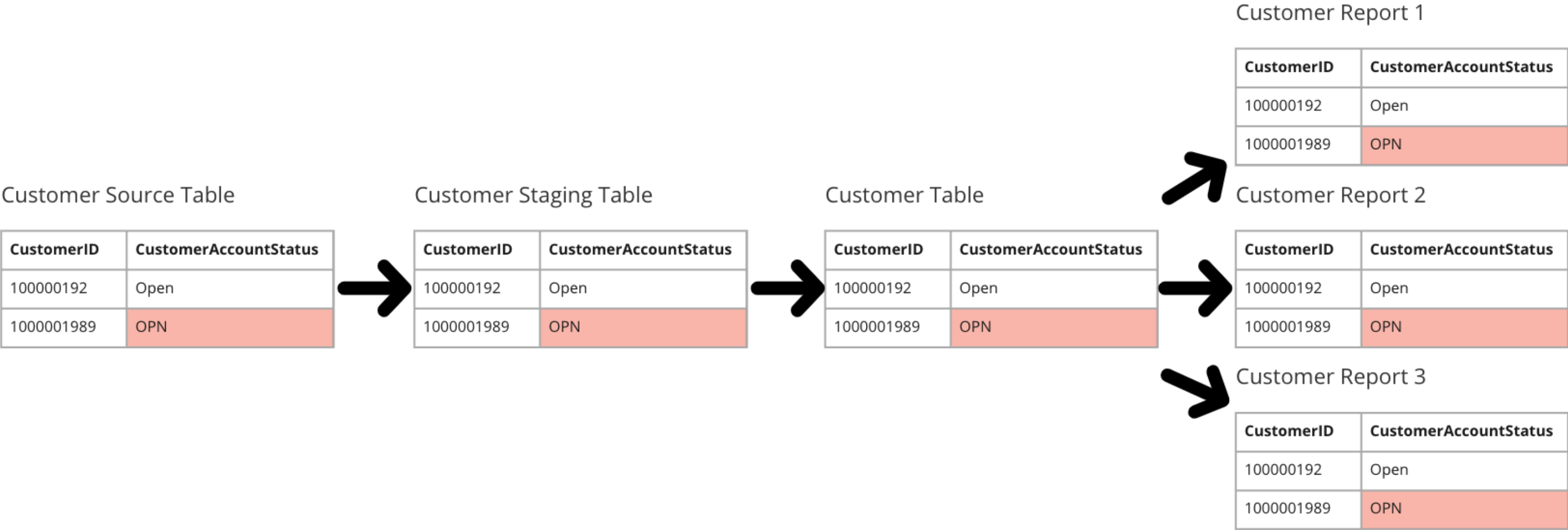
Customer
Table



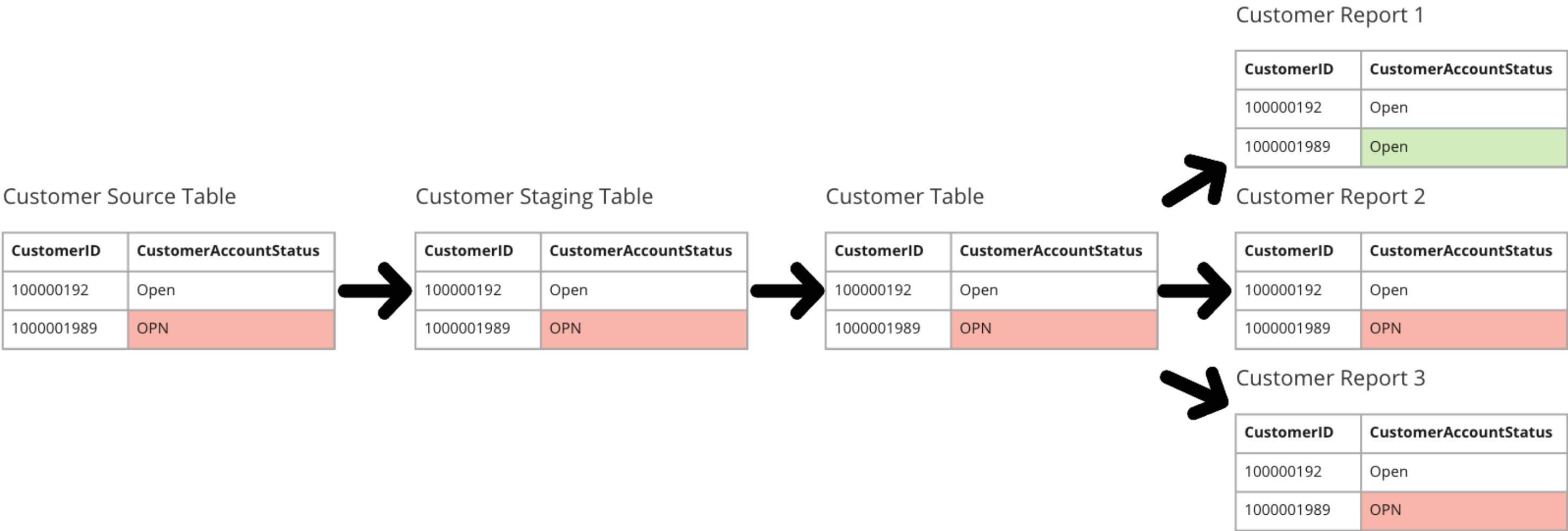
Customer Reports



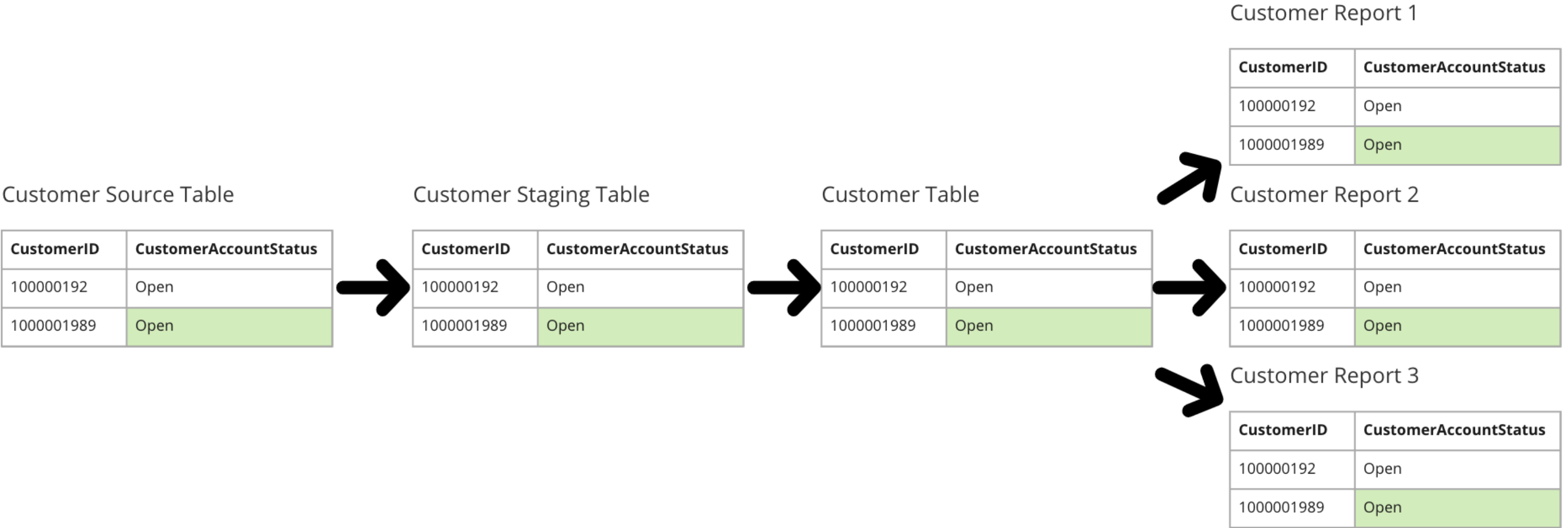
Metadata and data lineage example



Metadata and data lineage example bad practice



Metadata and data lineage examplebest practice



Let's practice!

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Data quality issues triage

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Using data lineage and metadata

Data Quality Process

1. Define data quality rules
2. Implement data quality rules
3. Monitor data quality rules for issues
4. Identify data quality issue
5. Triage data quality issue
6. Remediate data quality issue

... repeat!

Use data lineage and metadata to complete 5 + 6 with confidence

Step 1: Review the data profile

CustomerAccountType
Loan
Deposit
Credit Card
Deposit
Deposit

- Populated for 99% of records
- 1% are null
- 22% have a value of Loan
- 34% have a value of Deposit
- 30% have a value of Loan and Deposit
- 9% have a value of Credit Card
- 4% have other various values.



Data
Profile

Step 2: Identify where the rule is running

Data Quality Rule - All records must have a Customer Type with one of the following values: Loan, Deposit, Loan and Deposit, or Credit Card.

```
select *  
from CustomerSourceTable  
where CustomerAccountType <> ('Loan', 'Deposit', 'Loan and  
Deposit', 'Credit Card')
```

Customer Source Table

CustomerID	CustomerAccountType
100000192	Loan
100000198	NULL
100000123	Indep

Customer Staging Table

CustomerID	CustomerAccountType
100000192	Loan
1000001989	NULL
100000123	Indep

Customer Table

CustomerID	CustomerAccountType
100000192	Loan
1000001989	NULL
100000123	Indep

Customer Report 1

CustomerID	CustomerAccountType
100000192	Loan
1000001989	NULL
100000123	Indep

Customer Report 2

CustomerID	CustomerAccountType
100000192	Loan
1000001989	NULL
100000123	Indep

Step 3: Check the metadata

SEARCH: Customer Account Type

Metadata

Business Field Name: Customer Account Type

Business Definition: The type of account that the customer has or will open.

There are three main account type: Loan, Deposit, and Credit Card. Customers may open any of the three.

Data Owner: Rita Walker 

Last Updated Date: 01/01/2023

Technical Physical Field Name: CustomerAccountType

Technical Physical Table Name: CustomerSourceTable

Step 4: Correct the issue

Customer Source Table

CustomerID	CustomerAccountType
100000192	Loan
100000198	NULL
100000123	Indep

Outcome of Data Quality triage:

Update the data quality rule to allow for NULL values for a period of time. 

Correct the issues in the source and fix the bot's issue. 

Let's practice!

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